PHIL1002: Introduction to Critical Thinking

Lecture 2B: Argument Structure–I

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Agenda: Argument Structure

Part 1
1. The basics of standardising arguments
2. Tips and tricks

Part 2
1. Missing Premises
2. Revised Standard Form

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Basic standardising

1. Premiss
2. Premiss
So (or therefore, hence etc.)
3. Intermediate conclusion (sub-conclusion)
4. Premiss
So (or accordingly, in consequence etc.)
5. Main Conclusion

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The point of standardisation…

…is to get an argument into a clear form, where the direction of inference is reflected in the physical arrangement of the components on the page.

We do this by ensuring that **every inferred statement appears below every statement from which it is inferred**.

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No premise indicators…

This means that a standardised argument **never uses premise indicators.**

Any premise indicator words like 'because' will be replaced by conclusion indicator words like 'so'.

An argument which started out in the form 'C because P' would have to be converted to 'P so C'.

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An example

We never really know what is the case, as opposed to what merely seems to be the case; for, first, however certain we feel that we are right in our judgments, we could always be dreaming; and secondly, even our judgment about what is happening right here and now will have to be withdrawn if sufficient contrary evidence turns up in the future.

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How do we standardise this?

STEP ONE: Underline the inference indicators.

STEP TWO: Go through the passage trying to identify the separable statements and giving them each a letter of the alphabet.
You can separate the statements from each other by inserting angle brackets and then give a letter to each bracket.

STEP THREE: Put it in standard form.

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An example

We never really know what is the case, as opposed to what merely seems to be the case; for, first, however certain we feel that we are right in our judgments, we could always be dreaming; and secondly, even our judgment about what is happening right here and now will have to be withdrawn if sufficient contrary evidence turns up in the future.

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Step one…

*We never really know what is the case, as opposed to what merely seems to be the case; **for, first,** however certain we feel that we are right in our judgments, we could always be dreaming; and **secondly,** even our judgment about what is happening right here and now will have to be withdrawn if sufficient contrary evidence turns up in the future.*

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Step two…

(a) *<We never really know what is the case, as opposed to what merely seems to be the case>;* ***for, first, (b)*** *<however certain we feel that we are right in our judgments, we could always be dreaming>; and* ***secondly, (c)*** *<even our judgment about what is happening right here and now will have to be withdrawn if sufficient contrary evidence turns up in the future.>*

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Step three...

(b) However certain we feel that we are right in our judgements, we could always be dreaming

(c) Even our judgement about what is happening right here and now will have to be withdrawn if sufficient contrary evidence turns up in the future

Hence

(a) We never really know what is the case, as opposed to what merely seems to be the case

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Logical vs grammatical order...

The idea is that, as far as possible, you should write what is inferred **below** that which it is inferred from, so that the **main conclusion appears at the very end**.

In this example, this reverses the **grammatical** order of the original in favour of displaying most clearly its **logical** order.

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But be careful…

Govier's version of a standardised argument is different from the version given here.

She doesn't give you a **method** of standardising.

She just provides you with the standardised version and **then, after that,** numbers the statements in their final order. This numbering is pointless.

In this and the next lecture I will teach you a method of standardising–**Revised Standard Form.**

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Agenda: Argument Structure

Part 1
1. The basics of standardising arguments
2. Tips and tricks

Part 2
1. Missing Premises
2. Revised Standard Form

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Tips and tricks…

1. Missing inference indicators.

If it seems clear that one is missing, you can make it easier to determine the total structure by writing it in yourself, enclosing it in square brackets to show that you have added it.

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Tips and tricks…

2. The occurrence of modal words as inference indicators.

Easy to overlook because unlike normal indicators they **usually occur inside sentences** and hard to tell whether a modal word is functioning as inference indicator or in one of its other roles.

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Example…

John can't come out tonight. He is finishing his homework.

(b) John is finishing his homework

So

*(a) John **isn't** coming out tonight*

(i.e. John is able to come out tonight but has a good reason not to do so)

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Example…

John can't come out tonight. He is finishing his homework.

(b) John is finishing his homework

So

*(a) John **can't** come out tonight*

(i.e. John is unable to come out tonight; here there might be an implied suggestion that, say, his parents are depriving him of the opportunity)

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Tips and tricks…

3. Make sure you give disguised statements their own letter.

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Example…

The loss of B company means that we shall have to withdraw.

(a) <The loss of B company> <u>means that</u> (b) <we shall have to withdraw>

(a) B company has been lost

So

(b) We shall have to withdraw

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Tips and tricks…

4. Disguised statements often involve repetition, and repetitions should get the same letter.

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Example…

Dogs chase cats. It follows from this that they are stupid.

(a) <Dogs chase cats>. It follows from (a)<this> that (b)<they are stupid>.

Notice here, too, how the inference indicator is a set of words which are split into two by the disguised statement (a).

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Tips and tricks…

5. Inference indicators are whatever words are doing the job of indicating inferences.

6. Inference indicators can be split up.

7. Modal words acting as inference indicators should be replaced with non-modal words.

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Example…

(a)<When you depress the brake pedal you move the hydraulic fluid in the master cylinder>, and (b)<the pressure in an interconnected fluid system is the same in all parts of the system>, so (c)<the pressure you put on the pedal must be transmitted equally to the four wheel cylinders>.

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Example...

(a) When you depress the brake pedal you move the hydraulic fluid in the master cylinder.

(b) The pressure in an interconnected fluid system is the same in all parts of the system.

So

*(c) The pressure you put on the pedal **is** transmitted equally to the four wheel cylinders.*

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Tips and tricks...

8. Some compound statements can be split up but some have to remain as a single assertion.

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Example…

(b) However certain we feel that we are right in our judgements, we could always be dreaming

(c) Even our judgement about what is happening right here and now will have to be withdrawn if sufficient contrary evidence turns up in the future

Hence

(a) We never really know what is the case, as opposed to what merely seems to be the case

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Example…

(c) Even our judgement about what is happening right here and now will have to be withdrawn if sufficient contrary evidence turns up in the future

P: Even our judgment about what is happening right here and now will have to be withdrawn

Q: Sufficient contrary evidence turns up in the future

Why haven't I given these separate letters in my standardisation?

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Example…

(c) Even our judgement about what is happening right here and now will have to be withdrawn if sufficient contrary evidence turns up in the future

Statement (c) is a conditional.

'P, if Q' and 'If Q, then P'
mean the same thing.

Conditionals **cannot** be broken up into their separate constituents.

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Tips and tricks…

8. Some compound statements can be split up but some have to remain as a single assertion.

(i) Don't split conditionals

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Example…

(b) However certain we feel that we are right in our judgements, we could always be dreaming

(c) Even our judgement about what is happening right here and now will have to be withdrawn if sufficient contrary evidence turns up in the future

Hence

(a) We never really know what is the case, as opposed to what merely seems to be the case

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Example…

(b) However certain we feel that we are right in our judgements we could always be dreaming.

R: We feel certain that we are right in our judgments

And

S: We could always be dreaming

NO! It's another disguised conditional…

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Example…

(b) However certain we feel that we are right in our judgements we could always be dreaming.

R: *We feel certain that we are right in our judgments*

Look how I cheated—I dropped the word 'however' to get R. If I had left it in, it would look like this:

However certain we feel that we are right in our judgments

This doesn't assert anything in its own right.

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Tips and tricks…

9. Make sure that each item in an argument to which you give a letter in the process of standardisation is a complete assertion in its own right—no more and no less.

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For example…

… if I say in the course of an argument in the philosophy of religion:

Moreover, God can neither be created nor destroyed

it is obvious that the word 'moreover' should be excluded during standardisation because what follows it is a complete assertion in its own right.

The 'moreover' is superfluous.

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Tips and tricks…a qualification

Tip 8 needs qualification.
To see why, consider this example:

Owing to the incompatibility of the principle of universal causation and the idea that we have a genuine choice when we act, the only defensible theories on the question of free will and determinism are fatalism and libertarianism.

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Tips and tricks…a qualification

Tip 8 needs qualification.
To see why, consider this example:

Owing to (a) <the incompatibility of the principle of universal causation and the idea that we have a genuine choice when we act>, (b) <the only defensible theories on the question of free will and determinism are fatalism and libertarianism>.

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Tips and tricks…a qualification

(a) <the incompatibility of the principle of universal causation and the idea that we have a genuine choice when we act>

This is clearly not a statement, in its own right. It's a nominalised statement.

(a) The principle of universal causation is incompatible with the idea that we have a genuine choice when we act.

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Tips and tricks…a qualification

9. Make sure that each item in an argument to which you give a letter in the process of standardisation is a complete assertion in its own right–no more and no less,
or can be converted into such an assertion without change in meaning.

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Example…

There is a shadow that is spreading over the chimpanzee today: with the spread of agriculture and forestry, the habitat as well as the life of the chimpanzee is threatened. Forests are cleared to make way for cultivation and food trees are poisoned to leave space for better timber trees. Moreover, since chimps are susceptible to all the infectious diseases of man, wherever their populations are near new human settlements the apes are endangered by epidemics.

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Example…

There is a shadow that is spreading over the chimpanzee today: with the spread of agriculture and forestry, the habitat as well as the life of the chimpanzee is threatened. Forests are cleared to make way for cultivation and food trees are poisoned to leave space for better timber trees. Moreover, since chimps are susceptible to all the infectious diseases of man, wherever their populations are near new human settlements the apes are endangered by epidemics.

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Example…

(a)<There is a shadow that is spreading over the chimpanzee today>: (b)<with the spread of agriculture and forestry, the habitat as well as the life of the chimpanzee is threatened>. (c)<Forests are cleared to make way for cultivation and food trees are poisoned to leave space for better timber trees>. Moreover, since (d)<chimps are susceptible to all the infectious diseases of man>, (e)<wherever their populations are near new human settlements the apes are endangered by epidemics>.

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Example…

(a)<There is a shadow that is spreading over the chimpanzee today>: (b)<with the spread of agriculture and forestry, the habitat as well as the life of the chimpanzee is threatened>. ***(c)<Forests are cleared to make way for cultivation and food trees are poisoned to leave space for better timber trees>.*** *Moreover, <u>since</u> (d)<chimps are susceptible to all the infectious diseases of man>, (e)<wherever their populations are near new human settlements the apes are endangered by epidemics>.*

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Example…

(a)<There is a shadow that is spreading over the chimpanzee today>: (b)<with the spread of agriculture and forestry, the habitat as well as the life of the chimpanzee is threatened>. ***(c)<Forests are cleared to make way for cultivation> and (d) <food trees are poisoned to leave space for better timber trees>.*** *Moreover, <u>since</u> (e)<chimps are susceptible to all the infectious diseases of man>, (f)<wherever their populations are near new human settlements the apes are endangered by epidemics>.*

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Example…

(e) Chimps are susceptible to all the infectious diseases of man

So

(f) Wherever the populations of chimps are near new human settlements, these chimps are endangered by epidemics

(c) Forests are cleared to make way for cultivation

(d) Food trees are poisoned to leave space for better timber trees

So

(b) With the spread of agriculture and forestry, the habitat as well as the lives of chimpanzees are threatened

So

(a) There is a shadow that is spreading over the chimpanzee today

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Tips and tricks…

8. Some compound statements can be split up but some have to remain as a single assertion.

(i) Don't split conditionals
(ii) Conjunctions should be split

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Tips and tricks…

10. Pronouns should be replaced with nouns when you standardise

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Example…

(e) Chimps are susceptible to all the infectious diseases of man

So

*(f) Wherever **the populations of chimps** are near new human settlements, these chimps are endangered by epidemics*

(c) Forests are cleared to make way for cultivation

(d) Food trees are poisoned to leave space for better timber trees

So

(b) With the spread of agriculture and forestry, the habitat as well as the lives of chimpanzees are threatened

So

(a) There is a shadow that is spreading over the chimpanzee today

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Examples…

1. *The committee is going to deny the group permission to demonstrate because they **advocate** violence.*

2. *The committee is going to deny the group permission to demonstrate because they **fear** violence.*

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Avoid pronouns…

When you write something like 'from this we can see that..', it might be clear to **you** what you mean by 'this', but it might not be clear to anyone else.

Sometimes the context will sort this out.

But when we lose the grammatical clues of the immediate context—when we put an argument into standard form—the reference of the pronoun 'this' might be totally obscured.

To avoid this kind of problem, **avoid pronouns.**

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Tips and tricks…

1. Missing inference indicators.
2. The occurrence of modal words as inference indicators.
3. Make sure you give disguised statements their own letter.
4. Disguised statements often involve repetition, and repetitions should get the same letter.
5. Inference indicators are whatever words are doing the job of indicating inferences.
6. Inference indicators can be split up.
7. Modal words acting as inference indicators should be replaced with non-modal words.
8. Some compound statements can be split up but some have to remain as a single assertion.
   (i) Don't split conditionals
   (ii) Conjunctions should be split
9. Make sure that each item in an argument to which you give a letter in the process of standardisation is a complete assertion in its own right–no more and no less, *or can be converted into such an assertion without change in meaning*.
10. Pronouns should be replaced with nouns when you standardise

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Next lecture…

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